

Data documentation & metadata

Data Science

Week 1-DS-04

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Part-1 Data Provenance

1.1 Primary Source

Attribute	Detail
Survey Name	Adult Psychiatric Morbidity Survey 2014 (APMS 2014)
Publisher	NHS Digital / Health and Social Care Information Centre (HSCIC)
Survey Year	2014 — fieldwork conducted throughout 2014
Access Tier	Public access — aggregate statistical tables (OGL v3.0)
Sample Design	Stratified, multi-stage probability sampling
Total Sample	7,546 adults aged 16+ in private households across England
Chapters Used	9 chapters (ch-02 CMD · ch-04 PTSD · ch-05 Psychotic · ch-07 PD · ch-08 ADHD · ch-09 Bipolar · ch-10 Alcohol · ch-11 Drug · ch-12 Suicide)

1.2 Processing Chain

Raw NHS Excel files → prevalence extraction → Wilson Score CI computation → 702-row master dataset → hypothesis testing → distribution analysis.

1.3 File Inventory

File	Type	Task	Description
APMS2014_MH_Regional_Statistical_Model.xlsx	.xlsx	DS-01	Master dataset — 702 rows, 9 disorder groups, regional prevalence + Wilson CIs
APMS2014_MH_Statistical_Baseline_Model.ipynb	.ipynb	DS-01	Extraction, CI computation, composite burden scoring pipeline
Hypothesis_testing_setup.ipynb	.ipynb	DS-02	Chi-square and t-test implementation on CMD age-sex data
hypothesis_test_results.csv	.csv	DS-02	Results of 3 hypothesis tests with test statistics and p-values
apms_distribution_analysis.ipynb	.ipynb	DS-03	CIS-R band distributions, CMD by ethnicity, employment, region

2. Dataset Dictionary

2.1 Master Regional Dataset — APMS2014_MH_Regional_Statistical_Model.xlsx

702 rows × 10 columns. Covers 9 disorder groups, 26 disorders, 9 English regions, 3 sex categories.

Column	Type	Range / Values	Description
Chapter	String	ch-02 to ch-12	Source chapter file identifier
Disorder_Group	String	9 unique values	Disorder category (e.g., Common Mental Disorders, PTSD, Alcohol Use)
Disorder	String	26 unique values	Specific condition or screening measure (e.g., Any CMD, PTSD screen positive)
Sex	String	Men / Women / All adults	Sex category as recorded in the APMS survey
Region	String	9 GOE Regions	English Government Office Region
Prevalence_Observed_%	Float	0.20 – 84.49	Published weighted prevalence (%) from APMS tables
Prevalence_AgeStd_%	Float	0.20 – 84.49	Age-standardised prevalence; equals Observed where not separately published
Sample_n	Integer	155 – 14,579	Unweighted respondent count — denominator for Wilson CI calculation
CI_Lower_95_%	Float	0.07 – 82.24	Lower bound of 95% Wilson Score confidence interval
CI_Upper_95_%	Float	0.54 – 86.80	Upper bound of 95% Wilson Score confidence interval
CI_Width_pp	Float	0.24 – 7.26	CI width in percentage points (CI Upper – CI Lower)

2.2 Hypothesis Test Results — hypothesis_test_results.csv

3 test rows + 1 summary row.

Column	Type	Values / Notes
Test	String	Chi-square / t-test
Variables	String	Gender vs CMD · Age vs CMD · Male vs Female CMD Rates
Null Hypothesis	String	H0 statement for each test
p-value	Float / String	<0.001 (chi-square tests) · 0.0158 (t-test)
Conclusion	String	Reject H0 / Fail to reject H0 — all 3 rejected at $\alpha = 0.05$

2.3 Key Derived Variables

Variable	Derivation	Used In
CI_Lower / CI_Upper	Wilson Score formula: $(p + z^2/2n \pm z\sqrt{(p(1-p)/n + z^2/4n^2)}) / (1 + z^2/n)$, $z = 1.96$	Master dataset
CI_Width_pp	CI_Upper_95_% - CI_Lower_95_%	Master dataset
Composite Z-Score	For each region: average of standardised (z-score) prevalences across all disorders. Positive = above England average burden.	DS-01 burden ranking
Male/Female CMD Count	$CMD_ \% \times N / 100$ — converts published rates to observed counts for contingency tables	DS-02 chi-square tests

3. NHS Data Usage Compliance Checklist

APMS 2014 aggregate tables are published under the Open Government Licence v3.0 (OGL v3.0). No Data Access Agreement is required. The checklist below confirms compliance across all 4 areas.

#	Item	Requirement	Status
1	Data tier	Public access tables only — no individual/restricted data	✓ Compliant
2	No PII	No personally identifiable information accessed or retained	✓ Compliant
3	Ethics	Secondary analysis of published aggregates — no separate ethics approval required	✓ Compliant
4	Attribution	All outputs cite: APMS 2014, NHS Digital / HSCIC	✓ Compliant
5	Citation	McManus et al. (2016). Mental health and wellbeing in England: APMS 2014. NHS Digital.	✓ Compliant
6	No misrepresentation	Findings clearly dated 2014 — no claims of current prevalence	✓ Compliant
7	No endorsement	Reports marked as internal intern analysis — no NHS endorsement implied	✓ Compliant
8	Storage	All files held in secure internal project directory only	✓ Compliant
9	Cell suppression	Only cells published by NHS Digital used — no suppressed cells reconstructed	✓ Compliant
10	Permitted use	Research/learning purpose only — no commercial use or onward sale	✓ Compliant
11	OGL notice	OGL v3.0 notice reproduced in all shared documents	✓ Compliant
12	Reproducibility	Methods documented in Jupyter notebooks — Wilson Score formula verified	✓ Compliant

OVERALL: FULLY COMPLIANT — 12/12 items confirmed. No restricted data accessed; all outputs are aggregate statistics.

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4. Limitations

#	Limitation	Impact & Mitigation
L-01	Survey Year 2014	Estimates may not reflect post-2014 changes. All outputs clearly dated; no current-year projections made.
L-02	Private households only	Excludes institutionalised populations. Prevalence likely conservative for severe disorders (psychotic, bipolar).
L-03	Aggregate data only	Individual-level covariates cannot be controlled. Analysis is descriptive, not causal.
L-04	Binary gender	Survey records Male/Female only. Non-binary respondents not separately classified — limitation of source data.
L-05	Small cell sizes	Some disorder × region cells have $n < 300$ producing wide CIs. Wide-CI cells flagged with * in DS-01 regional tables.
L-06	Screening instruments	Some disorders measured by screening tools (AUDIT, ADHD screen, SAPAS), not clinical diagnosis. Names qualified accordingly.

5. Summary

Term	Definition
APMS 2014	Adult Psychiatric Morbidity Survey 2014 — nationally representative psychiatric survey of England adults aged 16+
CMD	Common Mental Disorders — includes GAD, depression, phobias, OCD, panic disorder, CMD-NOS
CIS-R	Clinical Interview Schedule Revised — structured diagnostic instrument; score ≥ 12 = clinically significant CMD
AUDIT	Alcohol Use Disorders Identification Test — AUDIT 16+ = high-risk drinking; AUDIT 20+ = probable dependence
SAPAS	Standardised Assessment of Personality – Abbreviated Scale — 8-item personality disorder screening tool
Wilson Score CI	Confidence interval method for proportions; outperforms Wald interval at extreme values and small samples
Composite Z-Score	Region-level index averaging standardised prevalences across all disorders; positive = above England average burden
OGL v3.0	Open Government Licence version 3.0 — permits free use and redistribution of public sector information with attribution
DAA	Data Access Agreement — required for restricted NHS data; not required for APMS 2014 aggregate tables
GOE Region	Government Office Region — 9 administrative regions of England used throughout APMS 2014